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Q1.(Basic Array Creation)

```
import numpy as np

arr1 = np.array([10, 20, 30, 40, 50])

arr2 = np.array([
    [1, 2, 3],
    [4, 5, 6],
    [7, 8, 9]
])

print("1D Array:")
print(arr1)

print("Shape:", arr1.shape)
print("Data Type:", arr1.dtype)

print("\n2D Array:")
print(arr2)

print("Shape:", arr2.shape)
print("Data Type:", arr2.dtype)
```

Q2.(np.zeros() and np.ones())

```
import numpy as np

arr1 = np.zeros(8)

arr2 = np.ones((4, 4))
```

```
arr3 = np.zeros((3, 3))  
print("a) 1D Array of 8 Zeros:")  
print(arr1)  
print("\nb) 4x4 Array of Ones:")  
print(arr2)  
print("\nc) 3x3 Matrix of Zeros:")  
print(arr3)
```

```
Q3.(np.arange())  
import numpy as np  
arr1 = np.arange(0, 21, 1)  
arr2 = np.arange(10, 51, 2)  
arr3 = np.arange(5, 101, 5)  
print("a) Numbers from 0 to 20:")  
print(arr1)  
print("\nb) Even numbers from 10 to 50:")  
print(arr2)  
print("\nc) Numbers from 5 to 100 (step 5):")  
print(arr3)
```

```
Q4.np.linspace()  
import numpy as np  
arr1 = np.linspace(0, 5, 10)
```

```
arr2 = np.linspace(-10, 10, 15)
print("a) 10 Equally Spaced Numbers between 0 and 5:")
print(arr1)
print("Length:", len(arr1))
print("\nb) 15 Equally Spaced Numbers between -10 and 10:")
print(arr2)
print("Length:", len(arr2))
```

Q5.Random Arrays)

```
import numpy as np
arr1 = np.random.rand(10)
arr2 = np.random.randn(3, 3)
arr3 = np.random.randint(10, 51, (4, 5))
print("a) 1D Array of 10 Random Numbers (0 to 1):")
print(arr1)
print("\nb) 3x3 Matrix of Random Numbers (Standard Normal Distribution):")
print(arr2)
print("\nc) 4x5 Array of Random Integers (10 to 50):")
print(arr3)
```

Q6.Vectors and Basic Operations)

```
import numpy as np
v1 = np.array([2, 4, 6, 8])
```

```
v2 = np.array([1, 3, 5, 7])  
addition = v1 + v2  
subtraction = v1 - v2  
multiplication = v1 * v2  
dot_product = np.dot(v1, v2)  
print("Vector 1:", v1)  
print("Vector 2:", v2)  
print("\nAddition:")  
print(addition)  
print("\nSubtraction:")  
print(subtraction)  
print("\nElement-wise Multiplication:")  
print(multiplication)  
print("\nDot Product:")  
print(dot_product)
```

Q7.(Matrices and Operations)

```
import numpy as np  
A = np.array([[1, 2, 3],  
              [4, 5, 6],  
              [7, 8, 9]])  
B = np.array([[9, 8, 7],  
              [6, 5, 4],  
              [3, 2, 1]])
```

```
addition = A + B

matrix_multiplication = A @ B

element_multiplication = A * B

print("Matrix A:")

print(A)

print("\nMatrix B:")

print(B)

print("\nMatrix Addition (A + B):")

print(addition)

print("\nMatrix Multiplication (A @ B):")

print(matrix_multiplication)

print("\nElement-wise Multiplication (A * B):")

print(element_multiplication)
```

Q8.Properties of Arrays)

```
import numpy as np

arr = np.random.randint(1, 101, (4, 4))

print("Random 4x4 Matrix:")

print(arr)

print("\nShape:", arr.shape)

print("Dimension (ndim):", arr.ndim)

print("Total Number of Elements (size):", arr.size)

print("Data Type:", arr.dtype)

print("Minimum Value:", arr.min())
```

```
print("Maximum Value:", arr.max())
```

Q9. Combined - Random + Reshape + Statistics)

```
import numpy as np
```

```
arr = np.random.randint(1, 51, 20)
```

```
matrix = arr.reshape(4, 5)
```

```
print("4x5 Matrix:")
```

```
print(matrix)
```

```
print("\nSum:", matrix.sum())
```

```
print("Mean:", matrix.mean())
```

```
print("Standard Deviation:", matrix.std())
```

```
print("\nMaximum Value in Each Row:")
```

```
print(matrix.max(axis=1))
```

Q10. (Mini Project - NumPy Application)

```
import numpy as np
```

```
try:
```

```
    # Ask user for number of random values
```

```
    n = int(input("How many numbers do you want to generate? "))
```

```
    if n <= 0:
```

```
        print("Please enter a positive number.")
```

```
    else:
```

```

arr = np.random.randint(10, 101, n)

print("\nGenerated Array:")

print(arr)

print("\nStatistics:")

print("Mean:", np.mean(arr))

print("Median:", np.median(arr))

print("Standard Deviation:", np.std(arr))

print("Minimum:", np.min(arr))

print("Maximum:", np.max(arr))

rows = int(np.sqrt(n))

if rows > 1 and n % rows == 0:

    cols = n // rows

    matrix = arr.reshape(rows, cols)

    print("\nReshaped 2D Array:")

    print(matrix)

    print("\nRow-wise Sum:")

    print(np.sum(matrix, axis=1))

else:

    print("\nCannot reshape into a proper 2D matrix.")

except ValueError:

    print("Invalid input! Please enter an integer.")

except Exception as e:

    print("Error:", e)

finally:

    print("\nProgram Finished

```